



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE**

GRADE 12

ENGINEERING GRAPHICS AND DESIGN P2

NOVEMBER 2024

MARKING GUIDELINES

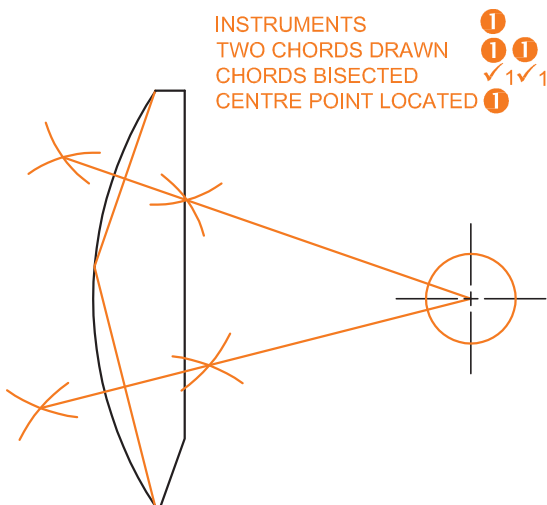
MARKS: 100

These marking guidelines consist of 7 pages.

ANSWERS		
1	MILLIMETRES / mm	1
2	21-05-2024	1
3	PIN VICE	1
4	023 685 1857	1
5	RIGHT VIEW	1
6	WOOD	1
7	4 : 5	1
8	5	1
9	80	1
10	<u>REVOLVED</u> SECTION	1
11	<u>SUCCESSIVE</u> SECTIONS	1
12	<u>HALF</u> SECTION	1
13	11	1
14	ADJUSTING CONE	1
15	C: M5 D: 18	2
16	E: 40°	1
17	MAXIMUM: 8,25	1
	MINIMUM: 7,7	1
18	See below	4
19		4
20		3
TOTAL		30

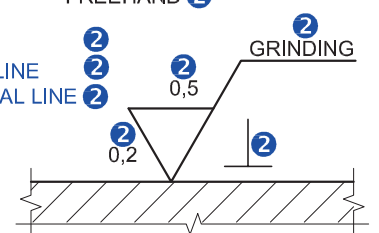
ANSWER 18

INSTRUMENTS
TWO CHORDS DRAWN
CHORDS BISECTED
CENTRE POINT LOCATED



ANSWER 19 FREEHAND 2

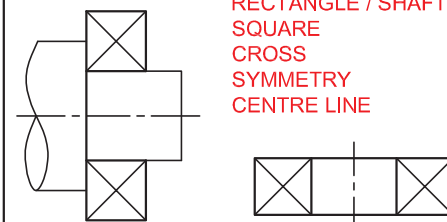
SYMBOL
TRIANGLE
INCLINED LINE
HORIZONTAL LINE



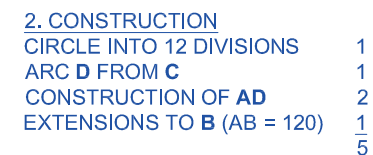
ANSWER 20

FREEHAND 3

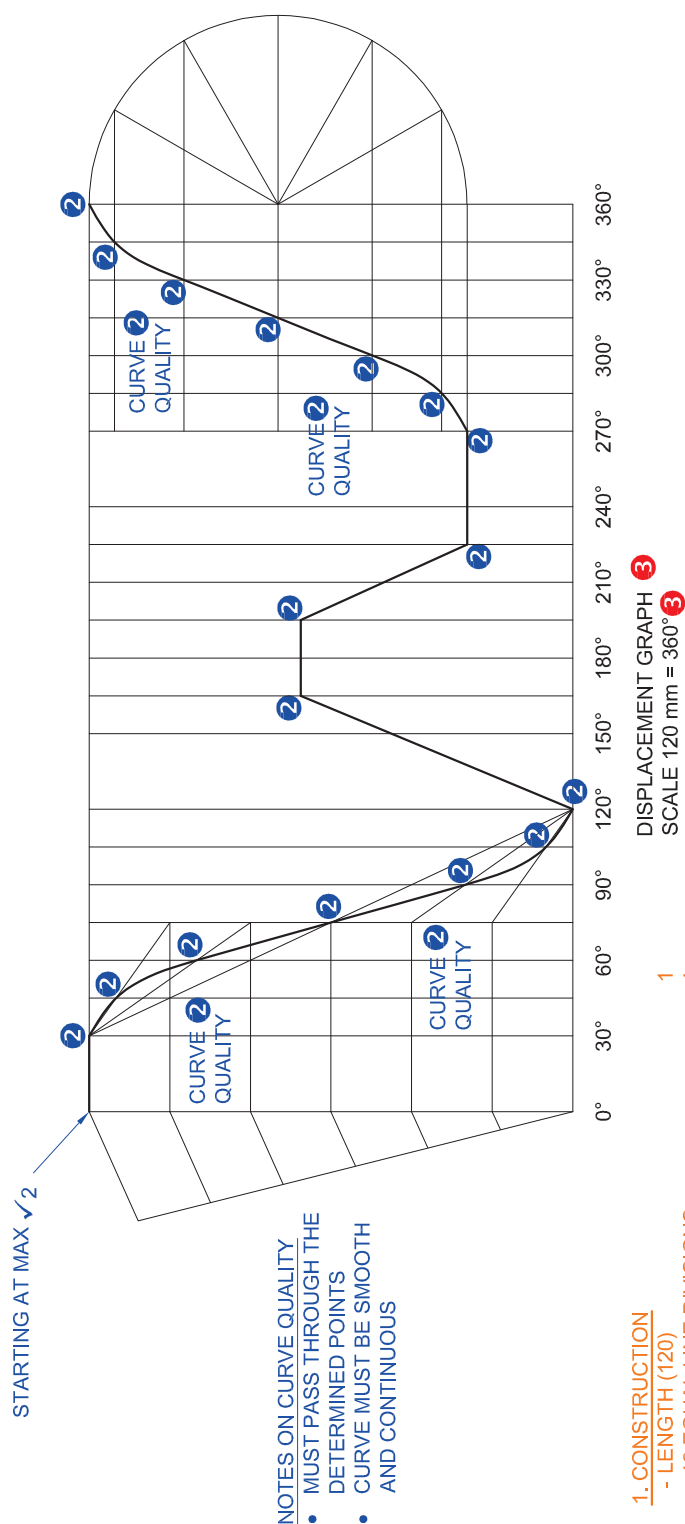
RECTANGLE / SHAFT
SQUARE
CROSS
SYMMETRY
CENTRE LINE



PAPER 2 QUESTION 1
GRADE 12
NOVEMBER 2024
MARKING GUIDELINES



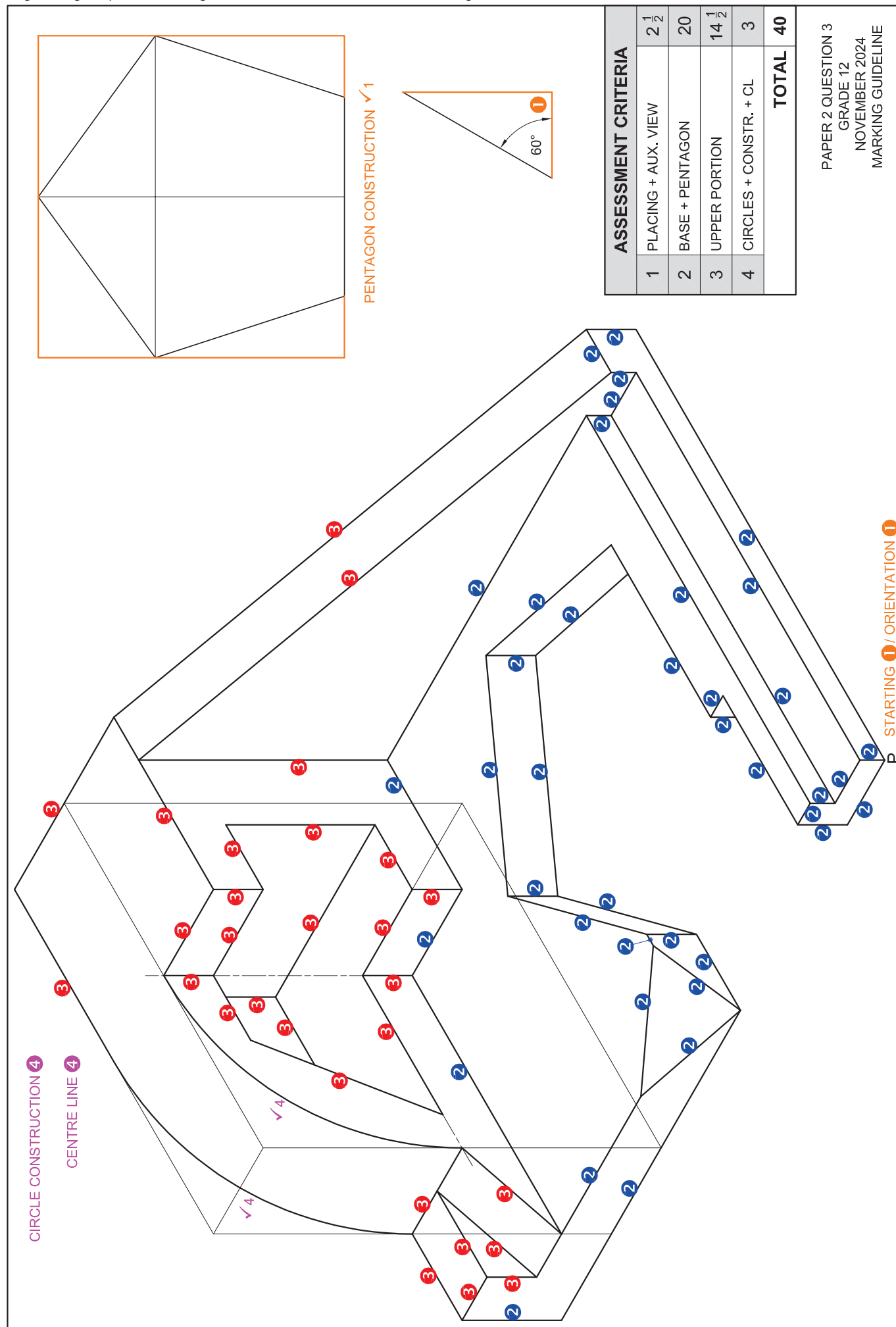
PAPER 2 QUESTION 2.1
GRADE 12
NOVEMBER 2024
MARKING GUIDELINES

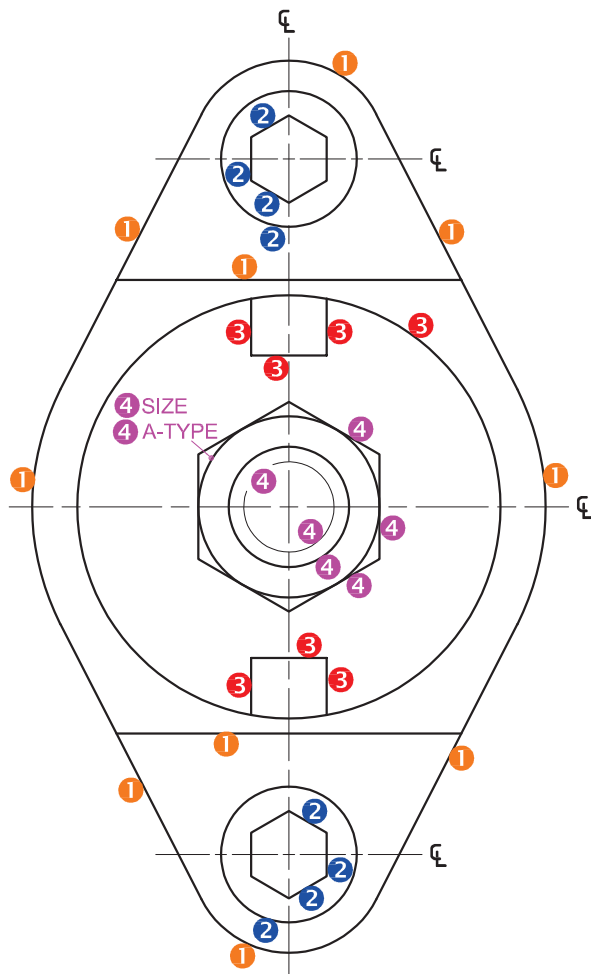


ASSESSMENT CRITERIA 2.2		
1	CONSTRUCTION	6
2	POINTS + CURVE	11 $\frac{1}{2}$
3	LABELS	1
SUBTOTAL 2.2		18 $\frac{1}{2}$

PAPER 2 QUESTION 2.2

PAPER 2 QUESTION 2.2
GRADE 12
NOVEMBER 2024
MARKING GUIDELINES





ASSESSMENT CRITERIA		
RIGHT VIEW		
1	RIGHT-END PLATE	5
2	BOLTS	4
3	COUPLING	3 ¹ / ₂
4	M16 NUT	4
SUBTOTAL		16 ¹ / ₂

PAPER 2 QUESTION 4
GRADE 12
NOVEMBER 202-
MARKING GUIDELINE

PISTON SEAL
MARK ALLOCATION

ROD GUIDE
ALTERNATIVE

ENLARGED BOLT
HEAD MARK
ALLOCATION
& ALTERNATIVE

ROD NOT SECTIONED

CENTRE LINE

26

10

1. CENTRE LINES ()
SANS COMPLIANT
CENTRE LINES (1 - 3 = 0.5; 4 - 6 = 1; 7 - 8 = 2)

2. ASSEMBLY
1 MARK FOR EVERY COMPONENT CORRECTLY
ASSEMBLED (11 PARTS - 1) = 10

ASSESSMENT CRITERIA	
HALF-SECTIONAL FRONT VIEW	
1	LEFT-END PLATE 10 ¹ / ₂
2	CYLINDER HOUSING 5
3	RIGHT-END PLATE 9
4	CONNECTING ROD + PISTON + SEAL 11
5	ROD GUIDE 3
6	COUPLING 6
7	BOLTS 14
8	M16 NUT 4
SUBTOTAL 62 ¹ / ₂	
GENERAL	
1	CENTRE LINES 3
2	ASSEMBLY 10
SUBTOTAL 13	

PAPER 2 QUESTION 4
GRADE 12
November 2024
MARKING GUIDELINE

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GRADE 12
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